FIITJEE

ALL INDIA TEST SERIES

FULL TEST – XI

JEE (Advanced)-2020

PAPER – 1

TEST DATE: 23-09-2020

Time Allotted: 3 Hours **Maximum Marks: 186**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-C.**

1. ***Section–A (01 – 04, 19 – 22, 37 – 40)*** *contains 12 multiple choice questions which out of 4 options have* ***only one correct answer****. Each question carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

   Section-A (05 – 12, 23 – 30, 41 – 48) *contains 24 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for all correct answer.*

   *Full Marks : **+4** If only (all) the four option(s) is (are) chosen.*
   *Partial Marks : **+3** If all the four options are correct but ONLY three options are chosen.*
   *Partial Marks : **+2** If three or more options are correct but ONLY two options are chosen and both of which are correct options.*
   *Partial Marks : **+1** If two or more options are correct but ONLY one option is chosen and it is a correct option.*
   *Zero Marks : **0** If none of the options is chosen (i.e. the question is unanswered).*
   *Negative Marks : **–1** In all other cases.*

2. ***Section-C (13 – 18, 31 – 36, 49 – 54)*** *contains 18 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

Physics

PART – I

SECTION – A
(One Options Correct Type)

This section contains **04 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

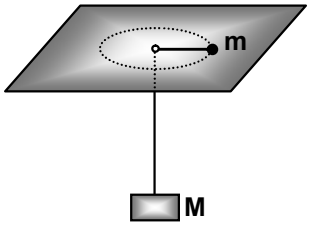
1. At t = 0, an object is projected from ground with an initial velocity of 100 m/s directed at angle 37° above the horizontal. At t = 4 sec it explodes in air into two parts with the mass ratio 1 : 2. Take the point of projection as origin and horizontal and vertical directions as x and y axes respectively. The motion of object is confined to the xy plane. At t = 10 sec, the coordinates of the lighter part is (400m, 60m). Find the coordinates of the heavier part at this moment. (i.e. t = 10 sec) (Given g = 10 m/s^2, $\cos 37° = \frac{4}{5}$)

(A) (500 m, 140 m)
(B) (500 m, 140 m)
(C) (500 m, 120 m)
(D) (1000 m, 120 m)

2. A pipe P open at one end but closed at the other end is filled with hydrogen gas at temperature 127°C. An another pipe Q open at both ends and having length equal to twice the length of pipe P is filled with a mixture of 1 mole of Hydrogen and 1 mole of Helium gases at temperature 147°C. The pipe P is vibrating in its first overtone and pipe Q is vibrating in its second overtone. The ratio of their frequencies of vibration $\frac{f_P}{f_Q}$ is

(A) $\sqrt{\frac{3}{5}}$
(B) $\sqrt{\frac{4}{3}}$
(C) $\sqrt{\frac{2}{3}}$
(D) $\sqrt{\frac{1}{3}}$

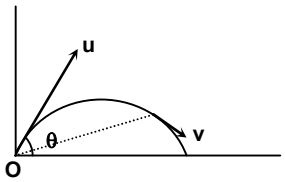
3. A particle of mass m is attached to one end of a light inextensible string which passes through a hole in a frictionless table top. At the other end of the string is attached a block of mass M. Initially the particle is moving in a circle of radius r with angular velocity ω. If the hanging block is pulled slightly downwards and then released, it will undergo small oscillation. Find the frequency of the oscillation.



(A) $\frac{1}{\pi}\sqrt{\frac{3m\omega^2}{M}}$
(B) $\frac{2}{\pi}\sqrt{\frac{3m\omega^2}{M}}$
(C) $\frac{1}{2\pi}\sqrt{\frac{3m\omega^2}{M}}$

(D) $\frac{3}{2\pi}\sqrt{\frac{3m\omega^2}{M}}$

4. A stone is projected in air directed at an angle θ above the horizontal. The maximum angle of projection θ with the horizontal so that the stone always moves away from the thrower during its motion in air, is (neglect air resistance)



(A) $\sin^{-1}\left(\frac{1}{3}\right)$

(B) $\sin^{-1}\left(\frac{2}{3}\right)$

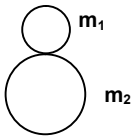
(C) $\sin^{-1}\left(\frac{\sqrt{8}}{3}\right)$

(D) $\sin^{-1}\left(\frac{\sqrt{5}}{3}\right)$

(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

5. Two elastic balls of masses m_1 and m_2 are placed on top of each other (with a small gap between them) and then dropped onto the ground. Then



(A) The upper ball ultimately receives the largest possible fraction of total energy if $\frac{m_2}{m_1} = 3$

(B) The upper ball will reach a maximum height of 4h if $\frac{m_1}{m_2} = 3$

(C) If $m_2 >> m_1$ the maximum height attained by ball m_1 after collision will be 9h.

(D) If $m_2 >> m_1$, the maximum height attained by the ball 2h.

6. Two insects each of mass m are at opposite ends of a diameter of a turntable of radius R. Moment of inertia of the turntable about its central vertical fixed axis is $I = \frac{2}{3}mR^2$. The turntable is rotating freely with angular speed ω and both the insects start crawling slowly towards each other on the turntable. Then

(A) The insect have to apply maximum horizontal force on the turntable when they are at a distance $\frac{R}{2}$ from the centre of the turntable.

(B) The insects have to apply maximum horizontal force on the turntable when they are at a distance $\frac{R}{3}$ from the centre of the turntable.

(C) The angular speed of the turntable when the insects exert maximum force on the turntable equal 3ω.

(D) The maximum force exerted by an insect on the turn table during the motion is equal to $3m\omega^2R$.

7. Consider a frictionless tunnel is made in Earth along a diameter. A uniform string of mass per unit length λ and length R/3 (where R is the radius of the earth) is suspended in the tunnel with its upper end at the surface of the earth. (Given acceleration due to gravity at the surface of the earth = g)

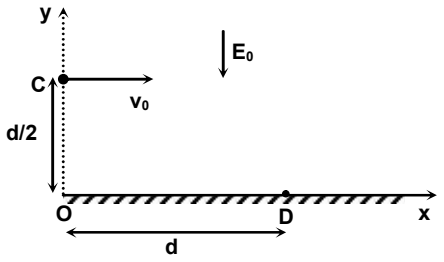
(A) The minimum force required to hold the string in the tunnel is $\frac{5\lambda gR}{18}$

(B) If the string is released in the tunnel from rest the speed of the string when its lower end touches the centre of the earth is $\sqrt{\frac{2gR}{3}}$

(C) If the string is released in the tunnel from rest the speed of the string when its lower end touches the centre of the earth is $\sqrt{\frac{gR}{3}}$

(D) The minimum work required to pull the chain completely to the surface of the earth is $\frac{2\lambda gR^2}{81}$

8. A particle of mass m and charge q moving in gravity free space with velocity v_0 parallel to x-axis enters a region of uniform electric field $\vec{E} = -E_0\hat{j}$ at point C. At the moment particle collides elastically with a smooth surface at D the electric field E is switched off and a magnetic field $\vec{B} = -B_0\hat{k}$ is switched on. If the particle hits the surface directly at point O after rebounding from point D, then which of the following option(s) is/are correct.



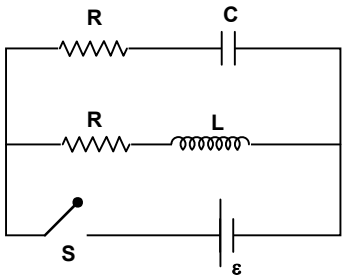
(A) $E_0 = \frac{mv_0^2}{qd}$

(B) $E_0 = \frac{mv_0^2}{2qd}$

(C) $B_0 = \frac{2mv_0}{qd}$

(D) $B_0 = \frac{mv_0}{qd}$

9. In the circuit shown in the figure initially the switch 'S' was open and capacitor uncharged. Now the switch 'S' is closed at time t = 0, then which of the following statement(s) is/are correct. (Given $C = \frac{L}{2R^2}$)



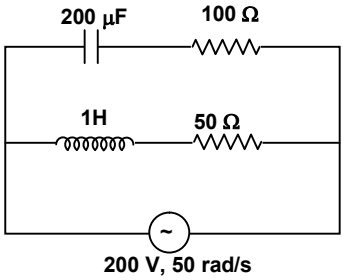
(A) The minimum current passing through the cell is $\frac{3\varepsilon}{4R}$.

(B) The current passing through the cell will be minimum at time t = RCℓn2.

(C) The charge on the capacitor when current passing through the cell is minimum, is q = $\frac{3C\varepsilon}{4}$

(D) When the current in the circuit is minimum, the ratio of potential difference across the inductor and the capacitor equals $\frac{2}{3}$.

10. In the given AC circuit as shown in the figure.



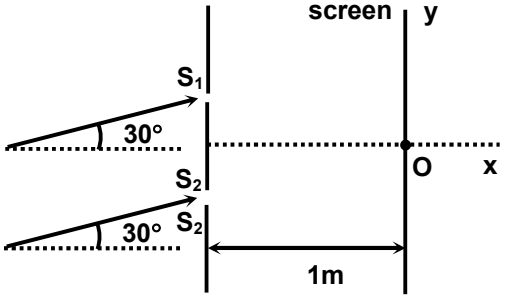
(A) The rms current through the ac mains is $\sqrt{10}$ A

(B) Total power dissipated in the circuit is 600 W

(C) The rms potential difference across the inductor is 100 V

(D) The phase difference between the currents passing in the capacitor and inductor is $\frac{\pi}{2}$

11. In a Young's double slit apparatus a parallel beam of light of wavelength λ = 0.3 mm with respect to air is incident on the slits at angle 30° with x-axis as shown in the figure. Distance of slits S_1 and S_2 from point O are equal. The medium to the left and right side of the plane of the slits have refractive indices $\frac{5}{3}$ and $\frac{25}{18}$ respectively. Distance between the slits S_1 and S_2 equals 1.5 mm. Distance between the plane of the slits and the screen is 1m. Then



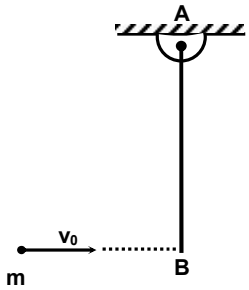
(A) The y-coordinate of the point on the screen where the phase difference between interfering waves is zero equals 0.75 m.

(B) The y-coordinate of the point on the screen where the phase difference between interfering wave is zero equals 0.36 m.

(C) If the intensity due to each light wave at point O is I_0 then the resultant intensity at point O equal $3I_0$.

(D) The y-coordinate of the nearest minima from the origin O is approximately y = –48 mm.

12. A thin uniform rod of length ℓ and mass 2m suspended vertically is free to rotate about a smooth horizontal axis which passes through its upper end A. A point mass m moving horizontally with velocity v_0 hits the lower end B of the rod and sticks to it. After collision the rod just reaches the horizontal position. Which of the following statement(s) is/are correct?



(A) The value of v_0 is $\sqrt{4g\ell}$

(B) The angular acceleration of the rod when it reaches the horizontal position is $\frac{9g}{8\ell}$

(C) The magnitude of vertical component reaction force by the hinge on the rod in horizontal position is $\frac{3mg}{5}$

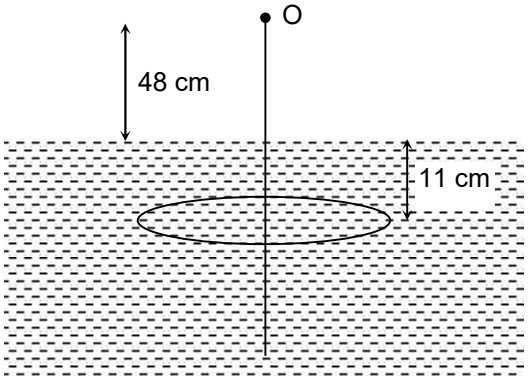
(D) The magnitude of acceleration of centre of mass of the system (rod + point mass) when the rod is in horizontal position is $\frac{4g}{5}$.

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

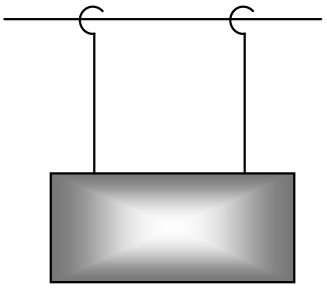
13. A soap bubble of radius R_1 is slowly given a charge q. Because of mutual repulsion of the surface charges, the radius increases slightly to R_2. The air pressure inside the bubble drops, because of the expansion, to $p\left(\frac{v_1}{v_2}\right)$, where p is the atmospheric pressure, v_1 is the initial volume and v_2 is the final volume. If $q = 4n\pi^2\varepsilon_0 PR_2(R_2^3 - R_1^3)$. Find the value of n. (neglect the surface tension).

14. A convex lens has a focal length 15 cm in air. The refractive index of lens is 1.5. The lens is placed inside the water at a depth 11 cm below the water surface. A point object is placed in air on the principal axis of the lens at a height 48 cm above the water surface as shown in the figure. The distance (in m) of final image of the object from the water surface, is



15. A non-uniform string of length $L = \frac{125}{64}m$ and mass m has a non-uniform linear density given by $\mu = \mu_0 x$ where x is distance from lower end of the string and μ_0 is a constant. Find the time required (in sec) for a transverse pulse generated at lower end of the string to travel to the other end of the string.

16. Consider a cylindrical body that can move without friction along a straight wire parallel to its axis of symmetry as shown in the figure. Particles moving horizontally at speed $v_0 = (3 + 2\sqrt{2})$ m/s collide the body uniformly from both left and right. Collisions with the left end of the body are elastic while those with the right end are completely inelastic though the particles do not stick to the cylinder after the collision. The speed (in m/s) of the cylinder after a long time will be (Assume that temperature of body remains constant)



17. The radiation emitted when an electron jumps from the third excited state to second excited state in doubly ionized Li^{++} ion (z = 3) falls on a metal surface to produce photoelectron, when the photoelectron with maximum kinetic energy is made to move perpendicular to a magnetic field of 2×10^{-4} T, its traces out a circular path of radius $\sqrt{\frac{9.1}{1.6}}$ cm. The work function of the metal (in eV) is (Given mass of electron = 9.11×10^{-31} kg)

18. A plane light wave of intensity I = 2kW/m^2 falls on a plane mirror. The intensity of reflected light is 60% of intensity of the incident light. The angle of incidence is 30°. If the pressure exerted by the light on the mirror is $n \times 10^{-6}$ N/m^2, find the value of n.

Chemistry

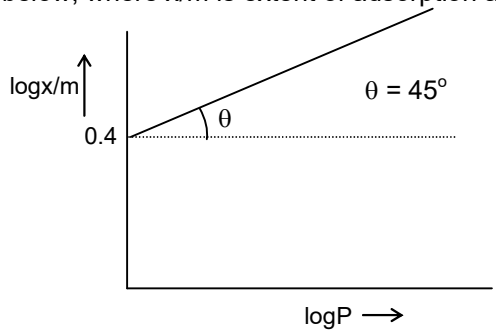
PART – II

SECTION – A
(One Options Correct Type)

This section contains **04 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

19. The purple colour produced in test of S^{2-} ion with sodium nitro prusside reagent is due to
(A) $Na_2[Fe(CN)_5NO]$
(B) $Na_4[Fe(CN)_5(NOS)]$
(C) $Na_4[Fe(CN)_6]$
(D) $Na_2[Fe(CN)_5(NOS)]$

20. According to Freudlisch's adsorption isotherm, the graph between log x/m and logP is given below, where x/m is extent of adsorption and P is the pressure



The extent of adsorption at pressure 2 atm is (Given log2 = 0.3 and log5 = 0.7)
(A) 8
(B) 2
(C) 5
(D) 9

21. A compound N_pS_q has cubic close packing (ccp) arrangement of S. N is present at alternative tetrahedral voids. If all the atoms along one body diagonal are removed. The empirical formula of the compound is:
(A) N_5S_{14}
(B) N_4S_5
(C) NS_2
(D) NS

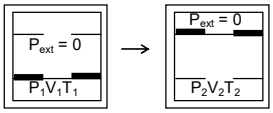
22. The number of angular nodes and radial nodes in $4d_{z^2}$ are respectively
(A) 0, 2
(B) 1, 1
(C) 2, 1
(D) 2, 0

(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

23. An ideal gas in thermally insulated container, at internal pressure P_1, volume V_1 and absolute temperature T_1 expands irreversibly against zero external pressure so that the pressure of gas becomes P_2, volume V_2 and temperature T_2. Which of the following statement(s) is/are correct?

(A) $\Delta S_{surrounding} = 0$

(B) $\Delta S_{system} = nR\ell n\frac{V_2}{V_1}$ 

(C) $\Delta H = 0$

(D) $P_1V_1^{\gamma} = P_2V_2^{\gamma}$

24. For a first order reaction $X(g) \longrightarrow 2Y(g) + Z(g)$, initially only 3 moles of X(g) was taken in 10 L rigid closed container at 300 K. After 10 seconds the reaction was stopped forcefully and a pinhole was made in the container at the same temperature. The molar ratio of X(g) and Y(g) gases coming out is 1 : 2. The correct option(s) is/are:
(Given molar masses of X, Y and Z are 16, 4 and 8 g/mole respectively $\ell n2 = 0.7$ and $\ell n3 = 1.1$, R = 0.08 atm L^{-1} K^{-1} mol^{-1})

(A) The total pressure inside the container when reaction was just stopped is 12 atm.

(B) The rate constant of reaction is 4×10^{-2} sec^{-1}.

(C) Moles of Z just after reaction was stopped forcefully is 1.0.

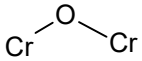
(D) The rate of formation of Y after 10 second is 0.4 mol L^{-1} sec^{-1}.

25. Fusion of chromite ore ($FeCr_2O_4$) with Na_2CO_3 in the presence of O_2 gives X which on acidification with conc. H_2SO_4 gives Y. Both X and Y contain chromium ion. Correct statement(s) is/are:

(A) Both X and Y are colourful compound and colour of these compounds is due to ligand to metal charge transfer.

(B) Both X and Y are paramagnetic in nature.

(C) Acidified solution of Y can convert Sn^{2+} ion into Sn^{4+} ion.

(D) Both X and Y have  bond..

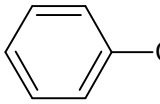
26. In an evacuated vessel of capacity 49.2 L, 4 moles of Ar(g) and 5 moles of $PCl_5(g)$ were introduced and the temperature is maintained at 240 K (assume that at this temperature decomposition of PCl_5 takes place). At equilibrium, the total pressure of the mixture was found to be 4.8 atm. (Given R = 0.082 atm L mol^{-1} K^{-1}). Correct statement(s) is/are

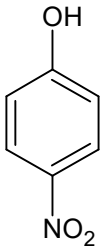
(A) The degree of dissociation of $PCl_5(g)$ into $PCl_3(g)$ and $Cl_2(g)$ is 0.6.

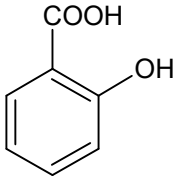
(B) K_P for the reaction $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ is 2.7 atm.

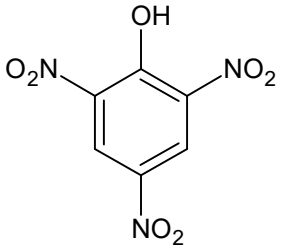
(C) At equilibrium total moles of gases are 12.

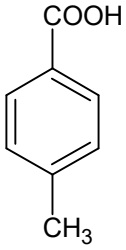
(D) On removing Ar(g) from the equilibrium mixture at constant pressure and temperature, the extent of dissociation of PCl_5 will increase.

27. Which of the following is/are stronger acid than  (benzoic acid)

(A) 

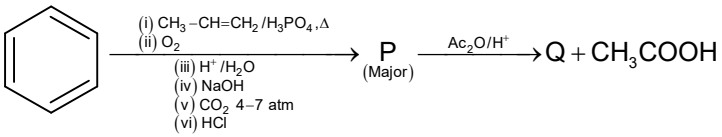
(B) 

(C) 

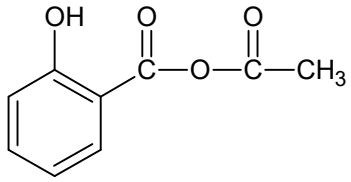
(D) 

28. Which of the following statement(s) is/are true?
(A) Sucrose on reaction with Fehling solution gives red precipitate.
(B) α-D-(-) fructofuranose and α-(L)-(+) fructofuranose are diasteomers.
(C) Glucose on reaction with nitric acid gives glucaric acid.
(D) In Lactose reducing half is β-D-glucose and non reducing half is β-D-Galactose.

29. Choose the correct option(s) for the following reaction sequence

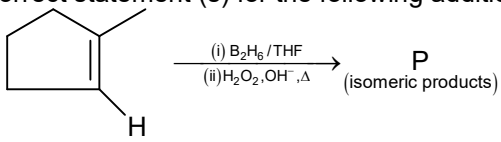


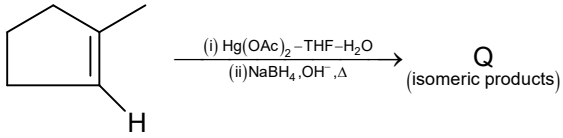
(A) Compound 'Q' has antipyretic action.
(B) Compound 'P' gives test with neutral ferric chloride.

(C) Compound 'Q' is 

(D) Compound 'P' can also be prepared by reaction of salicyaldehyde with Fehling solution followed by acidification.

30. The correct statement (s) for the following addition reactions is/are

(i) 

(ii) 

(A) Only one product is formed in (ii) reaction.
(B) Addition of water proceeds through syn-addition in both the reactions.
(C) Hydroboration in (i) reaction is controlled by steric factor.
(D) The product 'Q' is oxidized by CrO_3 while product 'P' is not oxididised by CrO_3.

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

31. The total numbers of stereoisomers obtained if NH_3 is replaced by bidentate en ligand in complex $\left[Co(CN)_2(NH_3)_2Cl_2\right]^-$ (en = $NH_2CH_2CH_2NH_2$)

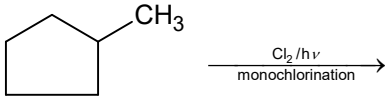
32. Calculate the value of [x – y] for $XeOF_4$ where x = total number of lone pair in $XeOF_4$, y = total number of $d\pi$ - $p\pi$ bonds in $XeOF_4$.

33. Among S_3O_9, P_4O_{10}, SO_2, SOF_3, SNF_4, N_2O_4, Al_2O_3 the total number of molecules containing $d\pi$ - $p\pi$ bond is

34. The solubility of AgCN(s) in 1 L of 0.4 M NH_3 is Y × 10^{-1} mol/L. The value of Y is…..
(Given that the value of K_{SP} AgCN = 5 × 10^{-10} and K_f of $[Ag(NH_3)_2]^+$ is 5 × 10^{10}

35. Melamine formaldehyde is a condensation polymer of melamine and formaldehyde. The number of nitrogen atoms in one molecule of melamine is X. X is…..

36. Total number of products (including both structural and stereoisomers) formed by monochlorination of the following compound is…….



Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **04 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

37. Let $x_1 = \sqrt[7]{2} + \sqrt[7]{8}$, $x_2 = \sqrt[7]{3} + \sqrt[7]{7}$ and $x_3 = \sqrt[7]{4} + \sqrt[7]{6}$, then
(A) $x_1 > x_2 > x_3$
(B) $x_1 < x_2 < x_3$
(C) $x_1 + x_3 < 2x_2$
(D) $x_1 + x_3 > 2x_2$

38. In a triangle ABC with sides a, b and c a semicircle touching the sides AC and BC is inscribed whose diameter lies on AB, then the radius of the semicircle is
(A) $\frac{a}{2}$
(B) $\frac{2abc}{s(a+b)}\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$
(C) $\frac{2\Delta}{a+b}$
(D) $\frac{2abc}{s(a+b)}\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}$

39. If $\int_0^\infty e^{-x^2}dx = a$, then $\int_0^\infty \frac{e^{-Jx}}{\sqrt{x}}dx$ where J > 0 is
(A) $2a\sqrt{J}$
(B) $\frac{a}{\sqrt{J}}$
(C) $a\sqrt{J}$
(D) $\frac{2a}{\sqrt{J}}$

40. Let A and B are two non singular matrices such that $B \neq I$, $A^5 = I$ and $AB^2 = BA$, then the least value of n for which $B^n = I$ is
(A) 63
(B) 64
(C) 31
(D) 32

(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

41. Considering the equation $\frac{2}{x}+\frac{5}{y}=\frac{1}{6}$ where x, y $\in$ N the number of solutions of the equation is
(A) 6 if both x and y are multiple of 4
(B) 6 if both x and y are the type 4n + 2, n $\geq$ 0
(C) 0 if x is multiple of 4 where y is of the form 4n + 2, n $\geq$ 0
(D) 0 if x is of the type 4n + 2, n $\geq$ 0 and y is the multiple of 4

42. If $\int \frac{9x^9-60}{5x^{13}+3x^{10}-5x}\,dx = -\ell n\left|5x^p+3x^q-5x^r\right|+c$ where p, q, r $\in$ I, then p + q + r is
(A) –3
(B) –5
(C) 3
(D) 15

43. Upon t – y plane the path defined by the equation $\frac{1}{x^{2m}}+\frac{1}{y^{2m}}=\frac{k}{(x+y)^n}=0$ is
(A) A parabola when $m=\frac{-1}{4}$, k = 1, n = 0
(B) A hyperbola when $m=\frac{1}{2}$, k = 1, n = 0
(C) A pair of lines of $m=\frac{1}{2}$, n = k = 1
(D) A pair of lines if $m=\frac{-1}{2}$, n = 1, k = 1

44. Which of the following statements are true?
(A) $n^2 - n + 41$ is a prime number for every n $\in$ N
(B) $\sum_{n=1}^{\infty}\frac{1+2^n}{n!}(\ln 2)^n$ is rational
(C) $\ln\left(\frac{1}{1\cdot 3}+\frac{1}{1\cdot 2\cdot 3\cdot 5}+\frac{1}{1\cdot 2\cdot 3\cdot 4\cdot 5\cdot 7}+.....\infty\right)$ can be simplified to an integer
(D) The coefficient of x^5 in $(1 - t)^{-6}$ is $^{10}C_6$

45. If $^{100}C_6+4\,{}^{100}C_7+6\,{}^{100}C_8+4\,{}^{100}C_9+{}^{100}C_{10}$ has the value equal to ^xC_y, then the value of (x + y) can be
(A) 114
(B) 115
(C) 198
(D) 199

46. Let $\bar{a}$ and $\bar{b}$ are unit vectors such that $|\bar{a}+\bar{b}|=1$. Let $\bar{\alpha}=\left(\left(\left((\bar{a}\times\bar{b})\times\bar{a}\right)\times\bar{a}\right)\times\bar{a}\right)\times\bar{b}$, then

(A) $|\bar{\alpha}|=\frac{\sqrt{3}}{4}$

(B) $\bar{\alpha}$ is perpendicular to $\bar{b}$

(C) $\bar{\alpha}$ is parallel to $\vec{a}$

(D) $\vec{\alpha}$ is parallel to $\vec{a}\times\vec{b}$

47. Let $f(x)=\begin{cases}\sin x & , \ -\pi\le x\le 0\\ \cos x & , \ 0<x\le\pi\end{cases}$ g(x) = f(|x|) + |f(x)| $\forall$ x $\in$ [–π, π], then

(A) g(x) = 0 has three real roots

(B) g(x) = 0 has infinitely many real roots

(C) $\lim\limits_{x\to 0} g(x)$ does not exist

(D) g(x) is a periodic with period π

48. The function of f(x) = max{(1 – x), (1 + x), 2}, x $\in$ (–∞, ∞) is

(A) continuous at all points

(B) differentiable at all points

(C) differentiable at all points except at x = 1 and x = –1

(D) continuous at all points except at x = 1 and x = –1

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

49. The least value of a for which the expression $4\,\text{cosec}\,x+\left(\frac{1}{1-\sin x}\right)-2a\left(\text{sgn}\left(\lim\limits_{n\to\infty}\left(\frac{1}{2}\right)^n\right)\right)$ has at least one solution in $\left(0,\frac{\pi}{2}\right)$

50. Total number of critical points in the domain of $f(x)=\frac{|2-x|}{x^2}$ are equal to

51. The equation of tangent to the curve y = 4 + $\sin^2$ x at x = π is y = k, then the value of k is

52. Let z_1, z_2 and z_3 be three non zero complex numbers such that $z_1 + z_2 + z_3 = z_1z_2 + z_2z_3 + z_3z_1 = 0$, then $\frac{|z_1||z_2|}{|z_3|^2}$ is equal to

53. Suppose x, $x^2 – 24$ and $x + x^2$ are consecutive terms of A.P., then the sum of all possible values of common differences of sequence is

54. A person flips 4 fair coins and discards those which turn up tails. He again flip the remaining coins and then discards those which turn up tails. If $P=\frac{m}{n}$ (m and n are coprimes) denotes the probability that he discards at least three coins, then the value of (m + n) is